# FLAT BASE CHANGE FOR SECTIONS OVER INTERSECTIONS OF QUASI-COMPACT OPENS

THE STACKS PROJECT AUTHORS

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*Allowed standard prerequisites (editorial).*

- Sections over a directed intersection of quasi-compact opens of a quasi-separated scheme equal the colimit of the sections over the members (more-groupoids chapter, equation-sections-of-intersection, carried as context span 3185-3196 with its directedness refinement). cited at proof line 3303. Computes Gamma(f^{-1}(E), O) and Gamma(E, O) as the colimits over the directed families f^{-1}(V_i) and V_i.
- Flat base change for cohomology of quasi-compact opens: for f : X to Y with the square cartesian and A to B flat, Gamma(f^{-1}(V), O_X) = Gamma(V, O_Y) tensor_A B for V quasi-compact open (Cohomology of Schemes chapter, Lemma coherent-lemma-flat-base-change-cohomology, canonical tag 02KH). cited at proof lines 3304-3305. Provides the single-open base case of the equality, which the colimit assembly extends to E.

*Source reference.* Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cited at proof lines 3304-3305: for each quasi-compact open V_i the flat base change isomorphism of section rings holds.

**Lemma 0.2.** *Suppose given a cartesian diagram*

$$\begin{array}{ccc} X & \longrightarrow & \mathrm{Spec}(B) \\ {\scriptstyle f}\downarrow & & \downarrow \\ Y & \longrightarrow & \mathrm{Spec}(A) \end{array}$$

*of schemes. Let $E \subset Y$ be an intersection of a nonempty family of quasi-compact opens of $Y$. Then*

$$\Gamma(f^{-1}(E), \mathcal{O}_X|_{f^{-1}(E)}) = \Gamma(E, \mathcal{O}_Y|_E) \otimes_A B$$

*provided $Y$ is quasi-separated and $A \to B$ is flat.*

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Let $X$ be a quasi-separated scheme. Let $E \subset X$ be a subset which is an intersection of a nonempty family of quasi-compact opens of $X$. Say $E = \bigcap_{i \in I} U_i$ with $U_i \subset X$ quasi-compact open and $I$ nonempty. By adding finite intersections we may assume that for $i, j \in I$ there exists a $k \in I$ with $U_k \subset U_i \cap U_j$. In this situation we have

$$\Gamma(E, \mathcal{F}|_E) = \operatorname{colim} \Gamma(U_i, \mathcal{F}|_{U_i}) \tag{0.2.1}$$

*Proof.* Write $E = \bigcap_{i \in I} V_i$ with $V_i \subset Y$ quasi-compact open. We may and do assume that for $i, j \in I$ there exists a $k \in I$ with $V_k \subset V_i \cap V_j$. Then we have similarly that $f^{-1}(E) = \bigcap_{i \in I} f^{-1}(V_i)$ in $X$. Thus the result follows from equation (0.2.1) and the corresponding result for $V_i$ and $f^{-1}(V_i)$ which is Cohomology of Schemes, Lemma 02KH. □